REVIEW

Role of Mineralocorticoid Receptors in Cardiovascular Disease

Nywe W.Y. Elton-Bott, Timothy J. Cole, Morag J. Young

ABSTRACT: Mineralocorticoid receptor (MR) antagonists (MRAs) are first-line therapy for primary aldosteronism and guideline-recommended treatment for heart failure (HF). The cardiovascular benefits of MRAs exceed blood pressure reduction, reflecting inhibition of inappropriate MR activation in cardiac and vascular tissues, and immune cells. Early preclinical studies demonstrated that aldosterone excess induces myocardial inflammation and fibrosis through MR-dependent pathways, and clinical trials established the efficacy of MRAs in reducing morbidity and mortality in HF. Nonsteroidal MRAs are now available, and aldosterone synthase inhibitors have commenced clinical testing; it is thus timely to revisit the mechanisms of MR activation in the heart and related cell types. This review will discuss key findings from preclinical and clinical studies of MRA use in HF and mineralocorticoid infusion, and update our understanding of MR actions in the normal myocardium and in the pathophysiology of primary aldosteronism and HF. Emerging data on aldosterone-mediated cardiovascular injury in patients with primary aldosteronism, including structural and functional cardiac changes and links to adverse outcomes, will be included, along with recent advances in MR-targeted pharmacology that offer improved selectivity and safety profiles. This review provides an updated understanding of MR-dependent pathways in the cardiovascular system and potential strategies for optimizing cardiovascular protection in both primary aldosteronism and HF.

Key Words: aldosterone ■ blood pressure ■ fibrosis ■ inflammation ■ morbidity

Cardiovascular disease (CVD) remains a leading cause of mortality, accounting for ≈24% of deaths in Australia.[1] Hypertension is the leading risk factor for CVD and remains a health care burden, with ≈1.4 billion people (aged 30–79 years) affected worldwide.[2] Despite advancements in antihypertensive therapies, blood pressure (BP) remains uncontrolled or resistant in many patients, suggesting an underlying secondary cause of hypertension. Primary aldosteronism (PA) has emerged as the most common and potentially curable form of secondary hypertension, contributing 3% to 13% of hypertension cases in primary care, up to 30% in specialized referral centers, and ≈20.9% of cases in patients with resistant hypertension.[3] PA was first described by Conn[4] and is characterized by renin-angiotensin-aldosterone system-independent aldosterone production due to an adenoma or hypertrophy of the adrenal zona glomerulosa.

Aldosterone synthesis is regulated by ANG II (angiotensin II),[5] serum potassium,[6] adrenocorticotropic hormone (ACTH),[5] and leptin[7] through induction of aldosterone synthase (CYP11B2). In patients with PA, aldosterone excess drives sustained mineralocorticoid receptor (MR) activation in target tissues, promoting sodium and water retention, volume expansion, hypertension with suppressed renin, hypokalemia, and metabolic alkalosis.[8] Inappropriate aldosterone-mediated MR activation in cardiomyocytes and other nonepithelial cells contributes to cardiovascular pathology both directly and indirectly (eg, via increased workload on the heart caused by increased BP, promoting hypertrophy).[9,10] The risks of stroke (2.58-fold), atrial fibrillation (3.52-fold), heart failure (HF; 2.05-fold), and coronary artery disease (1.77-fold) are all higher in patients with PA versus essential hypertension.[11] Current PA guidelines thus recommend that all individuals with newly diagnosed hypertension be screened to improve detection rates of PA and enable targeted treatment.[12]

This review will discuss the cellular mechanisms of tissue-inappropriate MR activation in the heart,

Correspondence to: Morag J. Young, Cardiovascular Endocrinology Laboratory group, Baker Heart and Diabetes Institute, 75 Commercial Rd, Melbourne, Victoria 3004, Australia, PO Box 6492, Melbourne, Victoria 3004, Australia. Email morag.young@baker.edu.au

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Nonstandard Abbreviations and Acronyms

ACTH	adrenocorticotropic hormone
AMPK	AMP-activated protein kinase
ANG II	angiotensin II
ApoE	apolipoprotein E
ASI	aldosterone synthase inhibitor
BMP	bone morphogenetic protein
BP	blood pressure
Ccl2	C-C motif chemokine ligand 2
CCR5	C-C chemokine receptor type 5
COX-2	cyclooxygenase-2
CREB	cAMP-response element binding protein
c-Src	cellular-Src
CTGF	connective tissue growth factor
CVD	cardiovascular disease
CXCL9	C-X-C motif chemokine ligand 9
DOC	deoxycorticosterone
EC	endothelial cell
ECM	extracellular matrix
ED	endothelial dysfunction
ENaC	epithelial sodium channel
eNOS	endothelial nitric oxide synthase
EPHESUS	Eplerenone Post-Acute Myocardial Infarction HF Efficacy and Survival Study
ERK1/2	extracellular signal-regulated kinase 1/2
Gal-3	galectin-3
HF	heart failure
HIF-1α	hypoxia-inducible factor-1α
HSD11β2	hydroxysteroid 11-beta dehydrogenase type 2
ICAM-1	intracellular adhesion molecule-1
IL-6	interleukin-6
iNOS	inducible nitric oxide synthase
IRAK	interleukin-1 receptor-associated kinase
KO	knockout
LDL	low-density lipoprotein
LV	left ventricular
MAPK/ERK	mitogen-activated protein kinase/extracellular signal-regulated kinase
MCP-1	monocyte chemoattractant protein-1
MMP2/9	matrix metalloproteinase-2 and 9
MR	mineralocorticoid receptor
MRA	mineralocorticoid receptor antagonist
MyD88	myeloid differentiation primary response 88
Myocyte-MR-KO	Cardiomyocyte-specific MR-KO mice
NF-κB	nuclear factor kappa B
NFAT	nuclear factor of activated T cell
NGAL	neutrophil-gelatinase-associated lipocalin
NO	nitric oxide
NR1D1/REVERBα	nuclear receptor subfamily 1, group D, member 1
PA	primary aldosteronism
PAI-1	plasminogen activator inhibitor-1
PKA	protein kinase A
RALES	Randomized Aldactone Evaluation Study
SATB2	special AT-rich sequence-binding protein 2
SGK1	serum- and glucocorticoid-regulated kinase 1
SK3	small-conductance calcium-activated potassium channel
TGFβ	transforming growth factor-β
TLR4	toll-like receptor 4
VCAM-1	vascular cell adhesion molecule-1
VSMC	vascular smooth muscle cell
WT	wild-type

integrating adrenal steroidogenesis, renin-angiotensin-aldosterone system control, and extra-renal MR signaling. Early preclinical and clinical evidence for the role of the MR in myocardial inflammation, fibrosis, and HF will be discussed alongside current data on the cardiovascular actions of aldosterone in patients with PA. Finally, we will provide an update on recent therapeutic developments, focusing on nonsteroidal MR antagonists (MRAs) and aldosterone synthase inhibitors (ASIs), to provide a rationale for MR-based intervention across PA and HF.

MINERALOCORTICOID RECEPTOR

The MR (encoded by NR3C2) is a ligand-activated transcription factor and member of the nuclear receptor superfamily that is essential for electrolyte homeostasis in epithelial cells.[13] Global MR knockout (KO) mice die shortly after birth from salt wasting, hyperkalemia, and dehydration, underscoring the fundamental role of the MR.[14] The MR also regulates cell function via rapid (within minutes), nontranscriptional mechanisms that include signal transduction pathways and protein-protein interactions with G protein-coupled receptors (ie, epidermal growth factor receptor).[15]

The MR binds both mineralocorticoids (aldosterone and deoxycorticosterone [DOC]) and glucocorticoids (cortisol in humans and corticosterone in rodents) with equivalent high affinity.[16] However, aldosterone-dependent control in epithelial tissues is conferred by high expression of HSD11β2 (hydroxysteroid 11-beta dehydrogenase type 2).[16] In contrast, HSD11β2 is absent or expressed at low levels in most nonepithelial tissues; thus, the MR predominantly responds to glucocorticoids, demonstrating ligand-specific regulation of the MR.[17] Our understanding of MR actions in nonepithelial tissues involves investigating inappropriate MR activation in cardiomyocytes, fibroblasts, endothelial cells (ECs), vascular smooth muscle cells (VSMCs), and immune cells, where cellular hypertrophy, and pro-inflammatory, profibrotic and oxidative stress (OxStress)-induced pathways are promoted.

Landmark clinical trials established spironolactone and eplerenone as disease-modifying therapies: the RALES (Randomized Aldactone Evaluation Study) reduced mortality and HF hospitalizations in severe HF with reduced ejection fraction,[18] and the EPHESUS (Eplerenone Post-Acute Myocardial Infarction HF Efficacy and Survival Study) improved survival and reduced sudden cardiac death following myocardial infarction with left ventricular (LV) dysfunction[19]; effects that are not solely attributable to MRA-mediated natriuresis but are consistent with antifibrotic and antiinflammatory actions observed in preclinical models. New nonsteroidal MRAs such as finerenone reduce cardiovascular events in patients with type 2 diabetes and chronic kidney disease, with improved receptor selectivity and tolerability, and distinct tissue distribution compared with steroidal MRAs.[20]

MR IN EPITHELIAL TISSUES

The kidney is a key target of mineralocorticoids, where aldosterone activates the MR in renal cells, including the principal cells of the distal nephron, to coordinate sodium, potassium, and fluid homeostasis. MR is also expressed in podocytes and glomerular ECs, where it modulates the integrity of the blood filtration barrier. The epithelial cells of the aldosterone-sensitive distal nephron (ie, late distal convoluted tubule, connecting tubule, and cortical collecting duct) modulate sodium reabsorption and potassium excretion via aldosterone-induced activation of the epithelial sodium channel (ENaC), sodium/potassium-adenosine triphosphatase pump, and renal outer medullary potassium channels. In addition, MR activation in intercalated cells maintains acid-base balance via hydrogen-potassium ATPases, potassium-chloride cotransporters, and the pendrin exchanger,[21] and modulates epithelial transport of electrolytes in the proximal tubule and loop of Henle, to indirectly regulate sodium transport. Together, these integrated actions control extracellular volume, electrolyte balance and BP.[22]

MR IN NONEPITHELIAL TISSUES

An increase in aldosterone can activate the MR to induce direct effects in nonepithelial tissues, including the heart, VMSCs, ECs, and other peripheral tissues, to restore homeostasis. However, inappropriate MR activation due to chronically elevated serum aldosterone levels can induce myocardial remodeling and dysfunction, with diffuse interstitial and perivascular fibrosis observed in the heart of rats treated with aldosterone infusion.[23] Preclinical studies of aldosterone excess have identified numerous mechanisms of MR-dependent myocardial inflammation and interstitial/perivascular fibrosis that are independent of renal MR actions.[9] Gene targeting in mice and other CVD models has revealed cell-specific and aldosterone-independent MR pathways that regulate disease mechanisms, as well as novel mechanisms that control cardiovascular homeostasis.

MR IN THE HEART

Under conditions of volume depletion, elevated serum aldosterone levels exert physiological chronotropic and ionotropic effects on the heart to increase cardiac output and restore homeostasis. These effects are driven by voltage-gated channels (ie, L-type, Cav1.2, and T-type, Cav3.2) that regulate the release and reuptake of calcium from the sarcoplasmic reticulum via induction of the ryanodine receptor 2 and phosphorylation of sarcoplasmic/endoplasmic reticulum Ca^{2+}-ATPase 2a (Figure 1). In vitro studies using rat cardiomyocytes identified a role for aldosterone in the maintenance of the cardiac sarcolemmal sodium/potassium pump and thus the resting membrane potential, which is critical for restoring membrane repolarization following each action potential.[24,25] Aldosterone induces an acute decrease in pump current, potentially due to a reduced affinity of the sodium/potassium-adenosine triphosphatase pump for intracellular sodium concentrations. Downstream effects include an increase in intracellular calcium levels, resulting in a positive inotropic effect. This is essential for normal heart function and adaptation to stress such as volume expansion.[25]

Mineralocorticoid-Dependent Models of Cardiac Remodeling

Preclinical models using aldosterone, DOC, or deoxycorticosterone acetate (a mineralocorticoid) infusion together

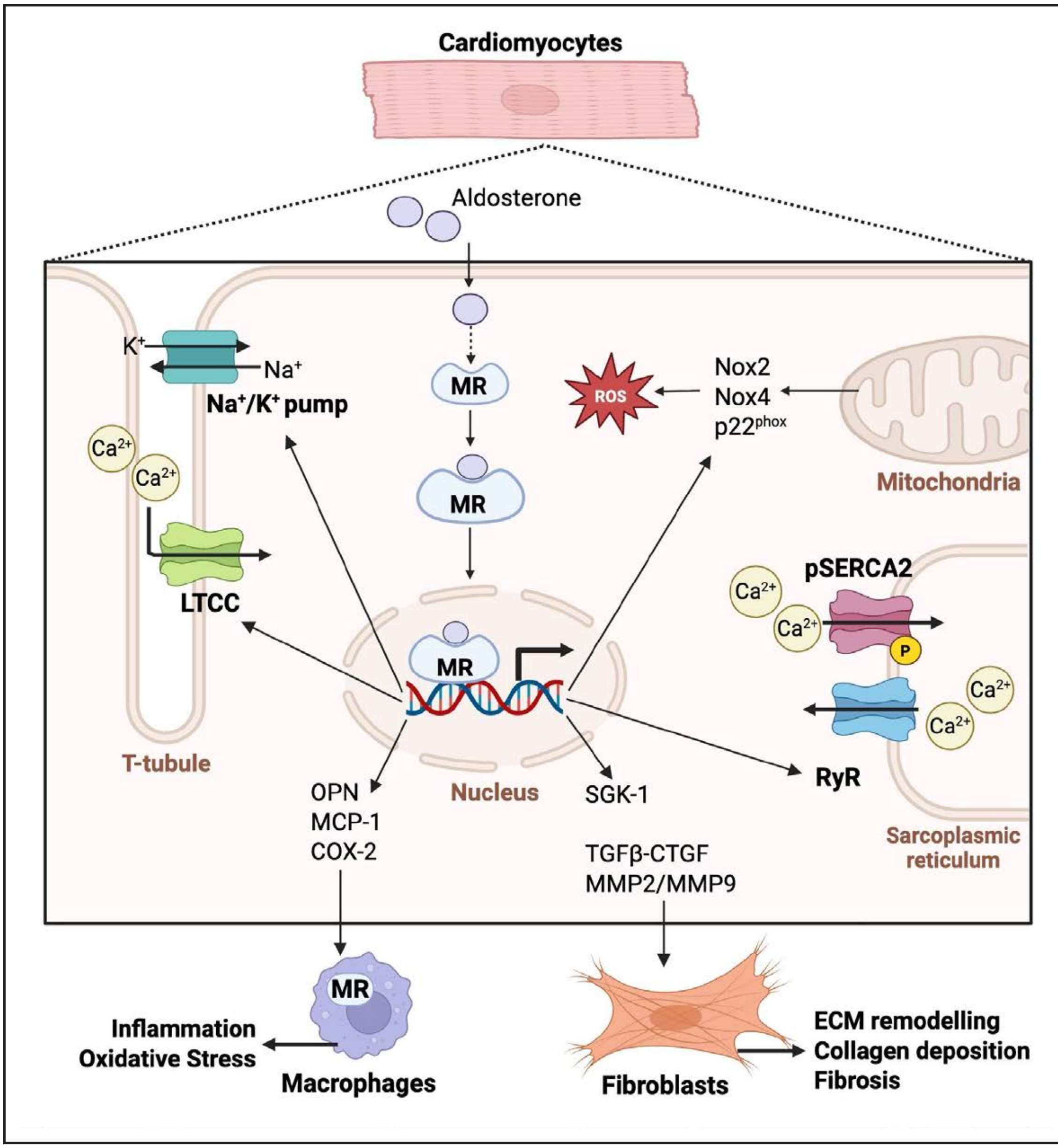

Figure 1. Mineralocorticoid receptor (MR) signaling in the heart has an important role in the pathogenic development of inflammation, oxidative stress and fibrosis.

MR activation by aldosterone promotes inflammation and fibrosis via transcriptional genomic actions, and calcium channel activation via MR-mediated signaling pathways. Ca^{2+} indicates calcium ions; COX-2, cyclooxygenase-2; CTGF, connective tissue growth factor; ECM, extracellular matrix; LTCC, L-type calcium channel Cav 1.2; MCP-1, monocyte chemoattractant protein-1; MMP2/MMP9, matrix metalloproteinase-2/9; A+/K+ pump, sodium/potassium pump; Nox2/4, NADPH oxidase 2/4; OPN, osteopontin; p22phox, human neutrophil cytochrome b light chain; pSERCA2, phosphorylated sarcoplasmic/endoplasmic reticulum calcium ATPase 2; ROS, reactive oxygen species; RyR, ryanodine receptor; SGK1, serum/glucocorticoid-regulated kinase 1; and TGFβ, transforming growth factor-beta. Created in BioRender. Young, M. (2026) https://BioRender.com/fibyomd.

with a high salt intake have been widely used to study MR-dependent cardiovascular injury.[26] An increased salt intake (via 1% NaCl drinking water) combined with nephrectomy induces sodium retention and facilitates a robust hypertensive response while suppressing the renin-angiotensin-aldosterone system, a feature seen in PA.[27] This model mimics chronic high salt intake common in humans (ie, >5 g/d of salt)[28] and contributes to MR-related damage by creating a pro-oxidative and proinflammatory[29] environment. For example, high salt intake can promote aldosterone-independent MR activation via Ras-related C3 botulinum toxin substrate 1-guanosine triphosphatase signaling.[30] Mineralocorticoid infusion has also been used in mouse models, where dosing regimens often differ from rats. These differences likely reflect the higher rate of metabolism in mice, which contributes to more rapid compound clearance.[31] When doses are standardized for body weight, aldosterone doses range from 3.85 to 30 µg/wk in mice,[32] while deoxycorticosterone acetate is commonly administered at ≈16.7 mg/wk,[33] with occasional higher doses reported (eg, 100 mg/wk).[34] Notably, longer study durations (≈8 weeks) tend to use slow-release formulations and a lower daily dose, versus shorter-term studies (≈3 weeks), which can lead to different tissue responses.

Uninephrectomized rats infused with aldosterone (0.75 µg/h) or DOC (20 mg/wk) over 8 weeks exhibited increased systolic BP and cardiac hypertrophy, with a 40% to 50% increase in LV mass.[9] These changes were shown to be independent of BP and cardiac hypertrophy, supporting direct MR-mediated remodeling of the heart.[9] Eight-week DOC/salt administration induced positive inotropy and systolic BP increases, which were absent in Myo-MR-KO mice, with lower atrial natriuretic peptide and B-type natriuretic peptide levels, suggesting reduced wall stress.[26] Similarly, in isolated rat hearts subjected to ischemia-reperfusion, aldosterone administration exacerbated cardiac injury, indicated by increased infarct area and apoptotic index. Spironolactone prevented these effects and reduced infarct size and apoptosis in adrenalectomized animals, indicating that MR blockade provides cardioprotection independently of circulating corticosteroids.[35] Partial protective effects were also observed with tempol (an antioxidant), indicating that ROS contributes to aldosterone-induced injury.[35] These findings show that DOC/salt treatment models activate pathophysiological MR similarly to aldosterone/salt and highlight the importance and breadth of chronic mineralocorticoid excess that drives maladaptive cardiac remodeling. Sex differences are also observed, with male mice showed higher sensitivity to deoxycorticosterone acetate-salt-related effects, with greater LV and renal hypertrophy at 6 weeks compared with females. Atrial natriuretic peptide, MCP-1 (monocyte chemoattractant protein-1), and TGFβ (transforming growth factor-β) expression were also increased in male, but not female, hearts.[36] Female patients also show a lower incidence of LV hypertrophy and renal disease before menopause compared with men, which is attributed to the cardioprotective effects of estrogen.[37] MR and estrogen receptors are members of the nuclear receptor family with similar structures and thus may provide shared protective effects in the same animal models.[37]

Cardiomyocyte-specific MR-KO mice (Myocyte-MR-KO) were used to dissect the direct contributions of MR signaling to the progression of cardiac pathology. At baseline, Myocyte-MR-KO hearts showed significantly lower expression of OxStress markers (NOX2, $p22^{phox}$), inflammatory mediators (CCR5 [C-C chemokine receptor type 5], MCP-1), circadian regulators (period homologue 2, NR1D1/REVERBα [nuclear receptor subfamily 1, group D, member 1], and profibrotic factors (TGFβ, CTGF [connective tissue growth factor], fibronectin, PAI-1 [plasminogen activator inhibitor-1]), indicating that MR-mediated activation of downstream targets may predispose cardiomyocytes for maladaptive responses. Myocyte-MR-KO mice were protected from deoxycorticosterone acetate/salt treatment and showed reduced collagen deposition, ECM remodeling, TGFβ1 expression, and activation of MMP2/9 (matrix metalloproteinases 2 and 9) activity and decorin[26] (Figure 1). Therefore, chronic aldosterone exposure regulates maladaptive hypertrophy and fibrosis via MR-dependent mechanisms, highlighting its pathophysiological impact on cardiac structure and function.

Regulation of the cardiac circadian clock has recently been described as a target of the MR, where its dysregulation may contribute to tissue dysfunction.[38] The molecular circadian clock is a cell-autonomous oscillator found in most cell types and is controlled via neuro-hormonal outputs, including the hypothalamic-pituitary-adrenal-axis regulation of cortisol.[26] Evaluation of the cardiac and endothelial transcriptome reveals regulation of core clock genes in DOC/salt- and aldosterone/salt-infused animals over 8 weeks, while acute studies of bolus administration of aldosterone given at lights-on and lights-off (12-hour interval) revealed both MR regulation of clock genes and time-of-day-dependent MR transcriptional activity, suggesting a reciprocal regulatory mechanism.[38] Specifically, MR activation regulates *Per1/2*, *Bmal, Cry2*, and *reverbα*, which have downstream effects on the timing of clock-controlled genes.[39] MR-KO also changes the expression of core clock genes, suggesting a fundamental role for the MR in regulating molecular clock timekeeping.[26,38]

Role for the MR in Other Models of Cardiac Inflammation and Fibrosis

Inappropriate activation of the MR has been characterized in numerous animal models of CVD.[9] Myocyte-MR-KO mice showed improved wound healing following myocardial infarction, attenuated progressive hypertrophy, and reduced ECM (extracellular matrix) remodeling of the noninfarcted myocardium, thereby protecting against HF progression.[40] Mechanistically, MR activation promotes SGK1 (serum- and glucocorticoid-regulated kinase 1), enhancing hypertrophic responses to mechanical and neurohumoral stress, and increasing OxStress via upregulation of Nox2 and Nox4.[41] Myocyte-MR-KO mice exhibited reduced SGK1 expression and enhanced ERK1/2 (extracellular signal-regulated kinase 1/2) activation, an intracellular signaling pathway associated with physiological hypertrophy (rather than pathological remodeling) and cardioprotection. Although these mice display mild cardiac hypertrophy at baseline in some studies, this feature may help reduce wall stress under increased afterload and delay maladaptive remodeling.[40] The transverse aortic constriction model further demonstrates cardiomyocyte MR signaling as a key driver of maladaptive cardiac remodeling during chronic pressure overload, independent of circulating aldosterone. Myocyte-MR-KO mice and fibroblast-specific MR-KO (MR^{COLCre}) mice were used to examine cell type-specific contributions. Myocyte-MR-KO mice were protected from LV dilatation and systolic dysfunction after 20 weeks of transverse aortic constriction, whereas MR^{COLCre} mice were not, highlighting the critical role of cardiomyocyte-expressed MR.[40] Together, these findings indicate a role for glucocorticoid-MR activation in cardiomyocytes in LV dilatation and dysfunction under pressure overload, providing a basis for MR-targeted therapies regardless of serum aldosterone levels.[41]

VASCULAR SMOOTH MUSCLE CELLS

Another extra-renal cell type targeted by aldosterone is the VSMC, which constitutes the middle layer (tunica media) of blood vessels, thus central to the regulation of vascular tone, BP, and blood flow distribution.[42] SMC-MR-KO mice have lower basal BP compared with MR-intact littermates and exhibit reduced resistance-vessel tone and vasoconstriction, implicating MR's role in peripheral vascular resistance and systemic BP regulation.[43] Aldosterone modulates VSMC contractility similarly to cardiomyocytes, where activation of L-type calcium channels initiates calcium influx, activates myosin light chain kinase, phosphorylates myosin light chains, and induces SMC contraction.[44] Moreover, Cav1.2 is a transcriptional target of aldosterone and is indirectly regulated via MR-mediated inhibition of microRNA-155 to prevent Cav1.2 mRNA degradation, leading to enhanced coronary vasoconstriction. In parallel, aldosterone-induced SMC-MR inhibits large-conductance Ca^{2+}-activated potassium channels to act as a negative feedback mechanism, closing L-type calcium channels and inhibiting contraction.[45] Other actions of SMC-MR activation include ROS production, which diminishes nitric oxide (NO) bioavailability and impairs soluble guanylyl cyclase activity, thereby reducing vasodilation capacity and facilitating vascular constriction.[46] ROS also upregulates MR expression via HIF-1α (hypoxia-inducible factor-1α)-dependent transcription, creating a feed-forward loop that amplifies MR activation. Importantly, vascular OxStress is ameliorated in SMC-MR-KO mice, demonstrating the receptor's role in physiological redox regulation.[47] Aldosterone also exerts rapid, nongenomic effects on intracellular signaling pathways, such as increased cAMP levels and phosphorylation of CREB (cAMP-response element binding protein). This response is mediated by calmodulin-sensitive adenylate cyclase, thereby linking cAMP and PKA (protein kinase A) signaling and rapid MR actions for aldosterone to influence vascular tone.[48]

MR activation in VSMCs regulates maladaptive vascular remodeling and fibrosis, as well as the pathogenic progression of atherosclerosis and myocardial infarction.[49] Following vascular injury caused by hypertension, diabetes, smoking, or mechanical interventions such as angioplasty or stenting, VSMCs undergo a phenotypic switch to a synthetic state.[49] In this state, VSMCs migrate, proliferate, and secrete ECM proteins to promote tissue remodeling processes that repair the vessel wall. Sustained activation drives pathological remodeling and vascular stiffening, thereby exacerbating ischemia and hypertension.[46] In vitro studies confirm that aldosterone actions on the SMC-MR directly promote VSMC proliferation and fibrosis via nongenomic pathways, including MAPK/ERK (mitogen-activated protein kinase/extracellular signal-regulated kinase), c-Src (cellular-Src), and Rho-kinase activation, which drive cytoskeletal reorganization and migration.[46] Genomic pathways, in contrast, upregulate profibrotic mediators such as CTGF, Gal-3 (galectin-3), and osteopontin, which stimulate ECM deposition.[50,51] SMC-MR-KO mice also exhibit attenuated vascular remodeling in response to aldosterone or ANG II.[52] Deletion of SMC-MR in aged male mice prevented fibrosis in the aorta, carotid, and renal arterioles via downregulation of CTGF, BMPs (bone morphogenetic proteins), and collagen expression. This protection was absent in females, highlighting sex-specific regulation of vascular remodeling by SMC-MR.[52]

MR activation upregulates BMP2 via transcriptional, OxStress, and inflammatory signaling pathways, driving osteogenic differentiation of VSMCs (Figure 2).[53] Mechanistically, MR activation represses microRNA-34b/c (normally suppresses osteogenic differentiation), thereby increasing SATB2 (special AT-rich sequence-binding protein 2), which promotes osteoblast gene expression.[54] Additional mechanisms include impaired autophagy and enhanced AMPK (AMP-activated protein kinase) signaling, which both reinforce the osteogenic phenotype.[55] In vivo studies further demonstrate that aldosterone infusion increases SMC transdifferentiation and calcium deposition in wild-type (WT) mice, while mice lacking SMC-MR are protected, exhibiting significantly reduced vascular medial and intimal mineralization.[46] Notably, pharmacological MR blockade prevents these changes, confirming SMC-MR as a critical driver of vascular calcification, contributing to arterial stiffening and elevated CVD risk.

VSMC-MR activation is a major driver of vascular inflammation, a process central to atherosclerosis and plaque instability. Aldosterone promotes MR-dependent transcription of cytokines, chemokines, and adhesion molecules, fostering leukocyte recruitment, OxStress, and endothelial dysfunction (ED). In vitro studies support a role for aldosterone-induced expression of proinflammatory mediators, including MCP-1, IL-6 (interleukin-6), and VCAM-1 (vascular cell adhesion molecule-1), via SMC-MR, establishing a chronic inflammatory state within the vessel wall.[56] Activation of MAPK/ERK, c-Src, and Rho-kinase pathways amplifies OxStress and promotes NF-κB (nuclear factor kappa B)-dependent transcription of

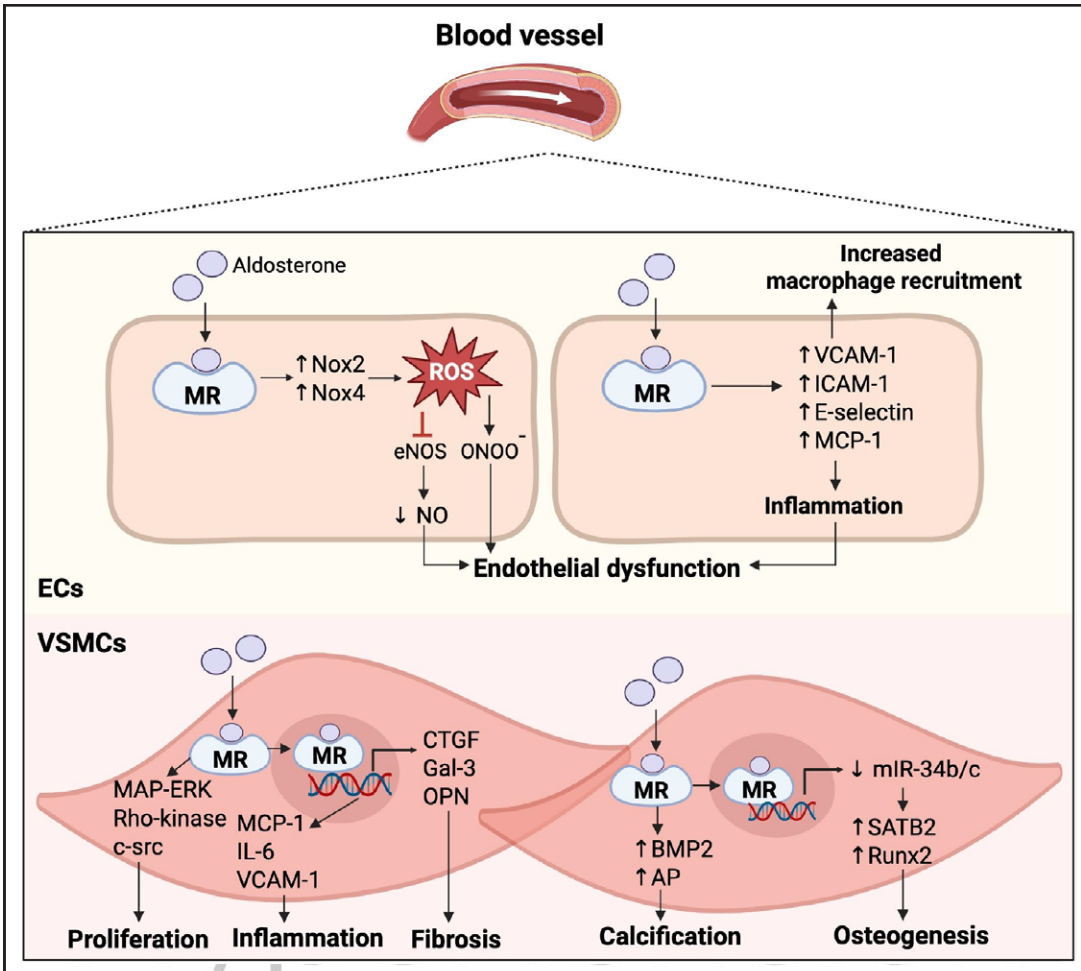

Figure 2. Mineralocorticoid receptor (MR) signaling drives a pathogenic response in blood vessel wall function specifically in endothelial cells and vascular smooth muscle cells.

MR activation induces endothelial dysfunction in endothelial cells (ECs) via increased NADPH oxidase activity and proinflammatory markers. Vascular smooth muscle cell (VSMC)-MR activate both genomic and nongenomic pathways, where aldosterone can induce vascular remodeling by inflammation, fibrosis and calcification. AP indicates alkaline phosphatase; BMP2, bone morphogenetic protein 2; c-src, cellular-src; CTGF, connective tissue growth factor; eNOS, endothelial nitric oxide synthase; Gal-3, galectin-3; ICAM-1, intracellular adhesion molecule-1; IL-6, interleukin-6; MAP-ERK, Mitogen-activated protein kinase/Extracellular signal-regulated kinase; MCP-1, monocyte chemoattractant protein-1; mIR-34b/c, microRNA-34b/c; NO, nitric oxide; Nox2/4, NADPH oxidase; ONOO-, peroxynitrite; OPN, osteopontin; ROS, reactive oxygen species; Runx2, runt-related transcription factor 2; SATB2, special AT-rich sequence-binding protein 2; and VCAM-1, vascular cell adhesion molecule-1. Created in BioRender. Young, M. (2026) https://BioRender.com/uq88jer.

inflammatory genes[47] (Figure 2). Aldosterone infusion in ApoE (apolipoprotein E)-KO mice enhanced atherosclerotic plaque inflammation, with increased lipid deposition and leukocyte recruitment. These effects were attenuated in SMC-MR-KO/ApoE mice, confirming that MR in VSMCs is necessary for this proinflammatory response. Importantly, these changes occurred independently of systemic BP, highlighting the cell-autonomous contribution of SMC-MR to vascular inflammation.

MR IN ECS

Aldosterone directly affects ECs to regulate expression and ion channel activity and cellular homeostasis.[57] MR activations increase ENaC expression, promoting sodium influx, cytoskeletal remodeling, and endothelial stiffening, which reduces NO bioavailability and impairs endothelial-dependent vasodilation.[58] In parallel, aldosterone modulates SK3 (small-conductance calcium-activated potassium channels) at the endothelial junctions, enhancing vasodilation via the hyperpolarization of adjacent VSMCs and ECs. SK3 channels sense calcium release in response to mechanical forces such as shear stress.[45] Evidence from SK3-KO mice demonstrates a role in BP regulation, as the loss of SK3 results in elevated systemic pressure.[59] Aldosterone activates endothelial NO synthase (eNOS) via the MR in ECs, promoting NO production essential for vasodilation and vascular homeostasis.[60] Rapid effects of aldosterone on eNOS activity have also been linked to the regulation of vascular contractility.[60] Under normal conditions, aldosterone preserves the endothelial glycocalyx, a layer of glycoproteins that covers the surface of the blood vessels and allows for the protection and proper functioning of the endothelium.[61] Therefore, by regulating ENaC and SK3 channels, activating eNOS, and preserving the glycocalyx, aldosterone plays a major role in vascular tone and BP regulation.

Dysregulated EC-MR activity is a significant regulator of vascular pathology through reduced NO bioavailability, increased OxStress mediators, and increased vascular reactivity. Importantly, the contribution of EC-MR to vascular function varies across vascular beds. In small mesenteric arteries, aldosterone impairs NO signaling even in EC-MR-KO mice, suggesting that dysfunction in these vessels is driven by MR in VSMCs or immune cells.[62] In contrast, aldosterone-induced ED, OxStress, and vascular stiffening in conduit arteries such as the aorta are MR dependent, occurring in WT but not EC-MR-KO mice. This highlights the central role of EC-MR in large vessels exposed to turbulent flow and shear stress, which, along with immune cell recruitment, promotes a proatherogenic phenotype.[63]

EC-MR-KO models first demonstrated that DOC/salt treatment induces maladaptive NO signaling in the vasculature. In WT mice exposed to DOC/salt, iNOS (inducible NO synthase) is upregulated, whereas eNOS is suppressed in the vessel wall, indicative of OxStress, NO dysregulation, and ED progression. In contrast, EC-MR-KO mice are protected from these vascular changes, with iNOS induction abolished and eNOS expression preserved.[64] Furthermore, N^G-nitro-L-arginine methyl ester treatment to inhibit NO attenuates vascular responses in WT, but not in EC-MR-KO mice, confirming that the EC-MR mediates pathological NO-dependent ED during mineralocorticoid excess.[65] In small resistance arteries, NO impairment is EC-MR independent, whereas in conduit arteries, EC-MR drives ED and vascular stiffening. Notably, females exhibited higher EC-MR expression and aldosterone sensitivity, particularly under sodium restriction, which elevated aldosterone levels.[66]

Complementary in vitro studies using human umbilical vein ECs are widely used as they retain key vascular function characteristics, including responsiveness to VEGF and NO production. Aldosterone decreases VEGF-induced eNOS phosphorylation and reduces NO production, resulting in impaired endothelial-dependent vasodilation.[67] Prolonged aldosterone exposure also increases cell adhesion molecule and chemokine expression, effects that are similarly blocked by MRAs and NF-κB inhibition. Moreover, excessive stimulation of the MR can contribute to glycocalyx damage, further impairing ED.[68]

Crosstalk between ECs and VSMCs is also regulated by MR-dependent effects. In ECs, aldosterone upregulates ENaC, increasing sodium influx and modifying cortical stiffness, which fine-tunes endothelial mechanics and NO availability to control vasodilation. In VSMCs, ENaC functions as a mechanosensor, translating mechanical stimuli into contractile responses to maintain myogenic tone. ENaC is expressed in both cell types and thus enables bidirectional communication and coordinated regulation of vascular tone.[69] MRAs can influence these interactions, underscoring the dynamic regulation of vascular function and homeostasis by aldosterone-mediated EC-VSMC crosstalk. The protection of female mice from these maladaptive responses by EC-MR-KO highlights an interaction among sex, diet, and MR signaling in the endothelium.[66]

MR IN INFLAMMATORY CELLS

Immune cells, specifically monocytes, macrophages, and T cells, express the MR, contributing to CVD, cancer, and metabolic syndrome.[70–72] Macrophages play a key role in the pathology of CVD, where bone marrow-derived monocytes are recruited to the heart and drive proinflammatory responses.[73] Macrophages adopt distinct functional phenotypes depending on microenvironmental signals. IFN-γ (interferon-γ), TNF-α, and other cytokines induced by tissue injury induce an M1 proinflammatory phenotype, leading to the secretion of cytokines, chemokines, and ROS.[74] In PA, MR activation can promote an M1 phenotype and chronic inflammation in the kidney and heart, as seen in preclinical models of aldosterone-salt loading.[75] MR-KO macrophages, in contrast, show an antiinflammatory phenotype, reduced macrophage accumulation in the heart, and improved myocardial structure in N^G-nitro-L-arginine methyl ester/ANG II-induced cardiac hypertrophy and fibrosis.[75] Myeloid-MR-KO mice are also protected from cardiac fibrosis and elevated systolic BP without a reduction in macrophage infiltration, which is seen in WT mice exposed to 8-week DOC/salt.[75] Myeloid-MR-KO mice showed improved cardiac function and neovascularization postmyocardial infarction, indicating an ability to enhance the cardiac repair process.[76] Moreover, T cells express the MR, which interacts with transcriptional factors NFAT (nuclear factor of activated T cells) and AP-1 to enhance inflammatory cytokine expression (ie, IFN-γ).[77] T-cell-specific MR-KO reduced cardiac hypertrophy, fibrosis, BP, and inflammation, with decreased accumulation of myeloid cells and lower cytokine expression compared with WT.[77] The therapeutic effects of MRAs in PA and HF are also linked to attenuation of macrophage activation and inhibition of inflammation,[74] thus highlighting macrophages as central mediators of aldosterone-induced inflammation.

The role of aldosterone-mediated inflammation has been demonstrated in rat cardiac fibroblasts, where MR activation increases NGAL (neutrophil-gelatinase-associated lipocalin) and Gal-3 expression.[78] MR-induced NGAL expression enhances M1 macrophage responses, whereas Gal-3 promotes myofibroblast proliferation and collagen deposition.[79] These mediators amplify TLR4 (toll-like receptor 4)-dependent signaling through recruitment of MyD88 (myeloid differentiation primary response 88) and activation of downstream IRAK (interleukin-1 receptor-associated kinase), leading to NF-κB signaling and contributing to cardiac remodeling and dysfunction. This inflammatory cascade further upregulated fibrotic (Col1a1, Col3a1) and inflammatory (Ccl2 [C-C motif chemokine ligand 2] and IL-6) markers[78] (Figure 3). In vitro studies using human monocytes treated with aldosterone suggest directional regulation of the OxStress subunit p22phox and the fibrinolysis inhibitor PAI-1, indicating transcription of a proinflammatory and profibrotic phenotype.[80] Moreover, proinflammatory cytokines released from macrophages, as observed in a DOC/salt model, include TNF-α, CXCL9 (C-X-C motif chemokine ligand 9), TGF-β, and MMP12.[81] Aldosterone infusion induces inflammatory cell recruitment and cytokine production in cardiovascular tissues, including osteopontin, MCP-1, COX-2 (cyclooxygenase-2), ICAM-1 (intracellular adhesion molecule-1), and VCAM-1.[82] Hence, the TLR4/

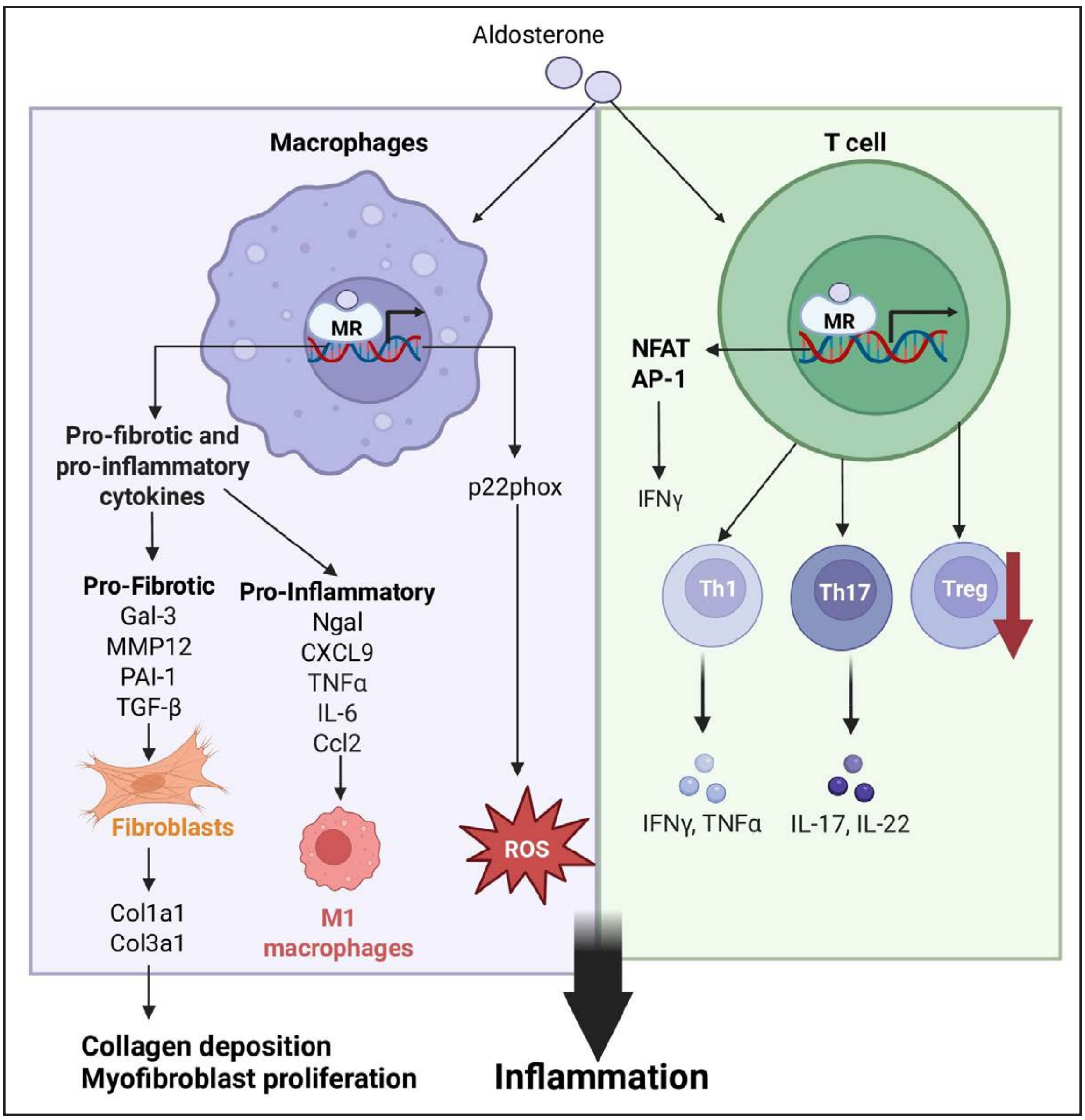

Figure 3. Mineralocorticoid receptor (MR) signaling in the macrophages and T-cells regulate tissue inflammation.
MR activation regulates the release of profibrotic and proinflammatory cytokines, promoting inflammation in other cells. AP-1 indicates activator protein-1; Ccl2, C-C motif chemokine ligand 2; Col1a1/Col3a1, collagen type I/III alpha 1 chain; CXCL9, C-X-C motif chemokine ligand 9; Gal-3, galectin-3; IFNγ, interferon-gamma; IL-6/IL-17/IL-22, interleukin-6/17/22; MMP12, matrix metalloproteinase-12; NFAT, nuclear factor of activated T cells; Ngal, neutrophil-gelatinase-associated lipocalin; PAI-1, plasminogen activator inhibitor-1; ROS, reactive oxygen species; T reg, Regulatory T cells; TGF-β, transforming growth factor-beta; TNF-α, tumor necrosis factor-alpha; and Th1/Th17, T helper cell 1/17. Created in BioRender. Young, M. (2026) https://BioRender.com/lgk9pio

NFκB pathway and OxStress provide a mechanistic link between aldosterone excess, macrophage activation and progression of cardiovascular injury.

Macrophage MR signaling has also been investigated in models of atherosclerosis, including LDL (low-density lipoprotein) receptor-deficient and ApoE-KO mice, because atherosclerosis underlies most CVD and is driven by leukocyte-mediated inflammation.[83] Monocytes recruited to sites of endothelial damage differentiate into macrophages and engulf oxidized LDL to form foam cells, thereby accelerating atherosclerotic formation. Oxidized LDL and cholesterol crystals act as damage-associated molecular patterns, activating TLR-dependent and NLRP3 inflammasome pathways that sustain proinflammatory cells to the lesion site.[84] Macrophage-specific MR-KO also reduces plaque formation and improves collagen content and overall plaque stability.[83] Moreover, eplerenone reduced the target-to-background ratio, a measure of arterial wall inflammation.[85] These findings highlight that excess aldosterone promotes inflammation in the heart and tissue remodeling through MR-regulated cell-specific pathways, driving end-organ damage.

The neutrophil-to-lymphocyte ratio, a well-established index of systemic inflammation reflecting the dynamic interplay between innate and adaptive immunity,[86] has emerged as a clinically relevant marker of cardiovascular and metabolic risk (reviewed in[87,88]). Notably, in patients with PA, an elevated neutrophil-to-lymphocyte ratio has been associated with both higher circulating aldosterone concentrations and the presence of chronic kidney disease,[89] suggesting that the neutrophil-to-lymphocyte ratio may serve as a monitoring option for cumulative inflammatory and fibrotic burden arising from sustained MR activation,[89] with implications in clinical practice for monitoring aldosterone-dependent target organ damage progression.

THERAPEUTIC REGULATORS OF THE MR AND ALDOSTERONE PRODUCTION

MR Antagonists

Pharmacological blockade of the MR, using agents collectively termed MRAs, is an effective therapy for CVD and, more recently, for chronic kidney disease, diabetes, and PA, with robust clinical trial evidence demonstrating reductions in morbidity and mortality.[18,90,91] Despite their proven efficacy, differences in molecular structure, receptor selectivity, and potency distinguish the classes of MRAs and define their clinical application.[92]

Steroidal MRAs, spironolactone and eplerenone, remain the most widely used agents in clinical practice and form the backbone of medical therapy for PA. Their advantages include extensive clinical experience, low cost, and strong efficacy in BP reduction and cardiovascular protection, particularly with spironolactone.[12] However, these benefits are offset by important limitations. Spironolactone's lack of receptor selectivity results in off-target androgen receptor antagonism and progesterone receptor agonism, leading to adverse effects such as gynecomastia, menstrual irregularities, and sexual dysfunction. Eplerenone improves MR selectivity and tolerability than spironolactone, but has lower potency, requiring higher doses to achieve comparable BP reduction.[19,93] An increased incidence of hyperkalemia is a common limitation of MRAs.[18] No trials have compared their antihypertensive equipotent doses; however titrated unequal dosing showed greater BP reduction with spironolactone, consistent with higher potency rather than superior efficacy.[93] Therefore, the current guidelines treat the agents as clinically interchangeable, but this is based on mechanism and titration rather than direct equipotent comparison. These tradeoffs among efficacy, selectivity, and safety have led to the development of alternative MR-targeted strategies.

Nonsteroidal MRAs are designed to overcome the pharmacological limitations of steroidal agents. Compounds such as finerenone and esaxerenone combine high MR selectivity with potent antagonism while largely avoiding off-target effects.[91] Their distinct binding modes alter MR coregulator recruitment and downstream transcription, resulting in stronger antifibrotic and antiinflammatory effects with reduced endocrine side effects.[94] Nonetheless, important limitations remain. Finerenone and esaxerenone are substantially more expensive and less widely available than classical MRAs. In contrast to spironolactone, available clinical trials and indirect comparisons suggest more modest BP reductions with finerenone, although these data are derived from studies using different populations, dosing regimens, and without equipotent head-to-head comparisons.[91,95] Long-term comparative data in PA are also limited, meaning that steroidal MRAs continue to dominate routine management despite the favorable pharmacological profile of these newer agents. Moreover, cortisol-mediated activation of the MR contributes to CVD; hence, MRAs may provide more complete blockade of dysregulated MR signaling than ASIs, potentially translating into greater cardioprotective effects in addition to BP reduction.

Aldosterone Synthase Inhibitors

ASIs represent a novel aldosterone-targeted strategy that suppresses hormone production upstream via selective CYP11B2 inhibition, rather than blocking the MR. This approach directly reduces both genomic and nongenomic effects of excess aldosterone and may prevent aldosterone escape seen with MRAs. ASIs are therefore particularly applicable to conditions driven by autonomous aldosterone excess, such as PA.[96] Early ASIs lacked selectivity and impaired cortisol synthesis via off-target CYP11B1 inhibition. However, second-generation

ASIs (eg, lorundrostat and baxdrostat) demonstrate high CYP11B2 selectivity and have shown promising BP-lowering effects in resistant hypertension and PA.[97,98] However, the role of ASIs in CVD is not yet established, although the cardiovascular and antiinflammatory benefits observed with MRAs suggest that ASIs may confer similar protective effects. However, important considerations remain; CYP11B2 inhibition alters steroidogenesis and may lead to accumulation of upstream precursors such as 11-deoxycorticosterone, which is increased by up to ≈700% in early studies.[99] Moreover, the long-term consequences remain unclear, with effects on renin-angiotensin-aldosterone system physiology requiring careful evaluation. Nevertheless, long-term efficacy, cardiovascular outcomes, renal safety, and comparative head-to-head trials of ASIs and MRAs remain to be determined. Future trials may also investigate combination therapies with SGLT2 inhibitors, which may mitigate hyperkalemia seen with ASIs, to clarify whether ASIs serve as a complementary therapy or an alternative to MRAs in aldosterone-mediated disease.[100]

CONCLUSIONS

Studies outlined in this review highlight the role of the MR in orchestrating aldosterone-dependent and aldosterone-independent pathways that shape cardiovascular health. Evidence from preclinical models, human physiology, and clinical trials has firmly established that inappropriate MR activation promotes cardiac hypertrophy, fibrosis, inflammation, OxStress, electrical instability, and adverse vascular remodeling. Importantly, these effects occur across cardiomyocytes, fibroblasts, ECs, VSMCs, and immune cells and highlight the MR as a central integrator of cellular stress responses in the heart. In PA, chronic aldosterone excess accelerates these MR-dependent mechanisms, explaining the disproportionate cardiovascular risk, including atrial fibrillation, coronary disease, stroke, and HF, even when compared with BP-matched essential hypertension. The advent of nonsteroidal MRAs and ASIs underscores a rapidly evolving therapeutic landscape, yet major gaps remain in translating MR biology into precision cardiovascular protection for patients with PA.

Future work must focus on several priority areas. First, deeper mechanistic mapping of cell type-specific MR signaling, including circadian regulation, nongenomic pathways, and MR-glucocorticoid receptor crosstalk, will refine our understanding of disease-causing processes. Second, improved biomarkers and imaging tools are needed to quantify MR activity and identify patients with early or subclinical mineralocorticoid excess, enabling more precise risk stratification and treatment selection. Third, emerging ASIs offer a promising strategy to directly suppress upstream mineralocorticoid production; hence, their efficacy and safety in conditions such as HF and other MR-dependent conditions require thorough evaluation in clinical trials. Finally, equitable implementation of screening for mineralocorticoid-dependent heart disease within hypertension management remains critical to preventing irreversible end-organ injury. Together, these directions underscore the potential to transform MR-targeted therapy from broad cardioprotection to truly mechanism-based precision medicine, reducing the substantial cardiovascular burden associated with dysregulated MR signaling.

ARTICLE INFORMATION

Affiliations

Cardiovascular Endocrinology Laboratory, Baker Heart and Diabetes Institute, Melbourne, Victoria, Australia (N.W.Y.E.-B., M.J.Y.). School of Translational Science, Monash University, Prahran, Victoria, Australia (N.W.Y.E.-B., M.J.Y.). Baker Department of Cardiometabolic Health, University of Melbourne, Victoria, Australia (M.J.Y.). Department of Biochemistry and Molecular Biology, Biomedicine Discovery Institute, Monash University, Clayton, Victoria, Australia (T.J.C.).

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